

Propositional logic – Introduction

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1 Formal theories

- We shall study *deductive* logic – the kind of reasoning we typically use in mathematics.
- In contrast to *inductive* logic – applied, for instance, in the physical sciences.

Focus is on *classical*, *axiomatic* deductive systems or *formal theories* used in mathematics.

- *proofs* that give rise to (mathematical) ‘theorems’ or ‘laws’.
So theorems are ‘deduced’, from some other statements
- Primitive statements – *axioms*
- Deductive Procedure

Example: Euclid’s theory of points and lines

Language of a formal theory

- *Alphabet* Σ – a non-empty countable set of symbols.
- Σ^* : set of all possible *strings* – finite sequences of symbols – on Σ

2 Propositional logic – PL

2.1 Alphabet

The alphabet of PL consists of

1. a non-empty countable set PV of *propositional variables*, denoted as $p_1, p_2, p_3, \dots$ (or simply, $p, q, r, \dots$),
2. *logical connectives* $\neg, \wedge, \vee, \rightarrow$, and
3. parentheses $(,)$.

2.2 Well-formed formulae (wffs)

The set $\mathcal{F}$ of *wffs* is the smallest set of strings over the alphabet given above, such that

- (i) any propositional variable in PV is a wff (i.e. a member of $\mathcal{F}$), called an *atomic wff*,
- (ii) if α is a wff, so is $(\neg\alpha)$,
- (iii) if α, β are wffs, so are $(\alpha \wedge \beta)$, $(\alpha \vee \beta)$, $(\alpha \rightarrow \beta)$.

“ m is positive, but not an integer”, would be translated as

“ m is not positive, unless n is negative”:

Abbreviation

- $\alpha \leftrightarrow \beta := (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$.

Proposition 2.1. The set $\mathcal{F}$ of all wffs of PL is countable.

Semantics of PL, that deals with truth and falsity of propositions.